IMA EDUCATIONAL Case Journal | Case Study

ima® The Association of Accountants and Financial Professionals in Business

ISSN 1940-204X

Chandler Corp: Choosing the Right Costing Method to Evaluate New Products

Paulette A. Ratliff-Miller
Associate Professor
Grand Valley State University

INTRODUCTION

Over many years, Chandler Corporation has become a profitable business by making awnings, pavilions, and other types of shade shelters. Most of the customers are municipalities looking for shelters for state parks or school districts that want covered walkways for students. Every job is custom, because each customer has different needs, and each location has different features. Chandler makes products when it receives an order; it does not produce products for inventory. While a customer might order multiple units for its site, the durability of the products means that the customer will not have to reorder replacements for decades.

So, when Ned Beale, the vice president of Sales and Marketing at Chandler, received a call from a swimming pool retailer in Florida that wanted a continuous supply of identical shelters, he was excited about the potential sales but a bit unsure as to whether Chandler could handle the order. The previous week he had gotten a similar call from a landscaping company inquiring about lattice-covered trellises the company wanted to incorporate into its landscape designs. He immediately texted Dan Smith, the CFO of Chandler, requesting a meeting to discuss whether to accept the new orders.

COMPANY BACKGROUND

Chandler Corporation, located in Chandler, Arizona, has been in the business of manufacturing fixed outdoor shade shelters for more than 30 years. Its customers are often public agencies (for example, city planners, schools, public parks, and so on), but they sometimes are landscape architects and designers who are developing outdoor areas for private customers (such as businesses, daycare centers, homes, and so on). The products are produced in a wide variety of configurations, including amphitheaters, pavilions, gazebos, trellises, transit shelters, walkway covers, dugout/bleacher covers, and fabric shade structures. These are all engineered to the customer's specification and are quick to install, which minimizes building time and disruption. Chandler provides a complete turnkey service from concept to completion with custom products for each site. The primary value driver is a quality design that can be easily installed on site.

PRODUCTS

Chandler specializes in shelters with tubular metal frames. Because of state and local building codes and site-specific constraints, Chandler's products are custom engineered and made for each location. A typical structure includes a steel frame, roofing, and other decorative and functional options. The frame consists of columns and supports for the roof,

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which can be painted in any color. Columns vary from simple designs, which use uprights consisting of one rectangular tube with flat plate anchors, to complex designs, which are created for visual appeal. Complex support structures can use multiple tubes for each support for aesthetics or strength. Customers might also choose a frame style, which can be connected to a stonework pedestal to achieve a more natural look.

The roofs on Chandler's shelters are typically made from metal sheets. Some products, such as trellises, have metal top panels with various cutout designs, which allow some light to shine through. Other products are designed to have fabric tops or sides attached. Working with designers and engineers, customers can customize their orders with optional accessories such as insulated ceilings, electrical outlet cutouts, gutters, solar panels, windscreens, integrated seating, ornamental railings, overhead gables, and cupolas.

SALES AND INSTALLATION

Chandler's products are distinctive and well-made, so it is not unusual for customers to seek out the company. Chandler does not perform any sort of market analysis and has no sales staff; instead it operates through independent representatives located throughout the country, who receive a percentage of the selling price as commission. These sales representatives work with a variety of manufacturers to provide all the products necessary for park and playground development. They also assist customers in finding independent, Chandler-trained installers to complete their construction on site.

PRODUCTION PROCESS

SALES ORDERS

The process starts when a customer who needs a shade shelter of some sort contacts a local sales representative. After discussing the customer's needs, the sales representative submits a preliminary design, referred to in the industry as a "napkin sketch," to Chandler whose engineers review it to determine whether the project can be manufactured in its facilities. The primary constraints are tube steel length limitations, which affect the size of the structure and affordability for the customer. If the project is feasible, then a contract price and conditions are established, and the customer decides whether the price is acceptable. Some contracts specify only one product, such as one amphitheater. Other contracts can be for multiple units, such as 50 roadside picnic table shelters. Some products are complex with many different parts, whereas other products have only a few parts.

PRODUCT DESIGN AND ENGINEERING

Once the contract is finalized, Chandler's drafting team will create more formal drawings of the shelter, including the lengths, angles, curves, slopes, and so on. Chandler engineers then examine the designs and determine the proper materials to be used in manufacturing. Because building codes vary considerably by location, designs must take into account not only global minimum building standards (International Building Code, IBC) but also local environmental conditions and regulations. For example, the California Division of State Architects (DSA) requires additional structural safeguards against earthquakes. A structure must be able to flex without breaking. The Florida Building Commission requires additional structure safeguards against hurricane-force winds. There the structure must be very rigid to withstand prolonged exposure to high winds. Thus, a basic steel square shade covering installed in California would be designed differently than a structure installed in Florida.

With custom design jobs, engineering time can be a limiting factor in the process. There are six engineers who work full time. Because of quality issues, these employees cannot practically work more than 50 to 55 hours a week. Adjusting engineering capacity is challenging because finding capable engineers is difficult, and it takes six months for an engineer to be fully functional for the company. Because engineers are costly, excess capacity in engineering is undesirable. Yet, without sufficient engineering hours available, delivery time can be extended for some projects. Unlike engineers, designers are more easily trained and readily available in the workforce. Chandler currently employs 10 designers who work 40 hours a week.

SCHEDULING, SETUP, AND FABRICATION

Scheduling is a critical function in Chandler's production process. In order for jobs to be completed in a timely fashion, materials or components must be scheduled to arrive or be finished and moved from one process just as they are needed in the next process. Chandler has found that scheduling costs rise in direct relation to the number of jobs started, as each requires the machines to be set up differently to handle the different components.

- **Schedule production.** Production is scheduled to meet the time constraints of the factory and the availability of the installer. Because the shelters are usually part of a larger project, the delivery of the shelter must be coordinated with the completion of the foundation. Installers do not want the product to sit at the site too long before it is installed, nor do they want to wait too long for the product.

- **Schedule labor.** Most manufacturing laborers are paid hourly and are cross-trained to work in different parts of the factory. There is currently sufficient capacity in the manufacturing facility to meet the production needs.
- **Materials.** Material orders must be scheduled to arrive as they are needed. Chandler does not have room at its facility to warehouse extra material. The primary material input is steel, which is usually ordered specifically for each job. The steel frame uses a variety of sizes and shapes of steel tubes. Most tubes are cut to the desired length in the factory. Some excessively wide tubes are ordered pre-cut. Other steel materials, such as plates and roofing, come in sheets and are cut to the desired length. Once delivered to the shop floor, materials are moved via overhead cranes and conveyer belts to the next process.
- **Cut and drill steel.** Given precise engineering designs, the workers carefully cut the steel tubes and sheets to the desired length and size. Some steel pieces will also have holes drilled into them for assembly on site and attachment to the foundation. There can be many parts and cuts for each product, and each must be cut perfectly the first time to avoid waste of critical material. When customers order multiple units of the same kind, the same cuts can be made multiple times, reducing the cutting and drilling machine setups needed.
- **Weld parts.** Some parts are welded together—for example, plates are welded onto the ends of support columns, which are used to anchor the tube to the foundation. The company uses an internal bolting system, and thus, nuts or bolts are frequently welded to the frame. Once the sections of steel are cut, drilled, and welded, they are sand blasted to make the metal smooth and ready for painting.

POWDER COATING

Powder coating is a painting process that applies a free-flowing, dry powder to metal surfaces, which does not require a solvent to keep the binder and filler parts in a liquid suspension form. The coating is applied electrostatically and is then cured under heat to allow it to flow and to create a "skin" over the metal surface. Because powder coating does not have a liquid carrier, it can thickly and evenly coat both horizontal and vertical surfaces. Multiple colors can be applied before curing to achieve the desired color and texture. Paint colors are easily changed with little cost. Chandler currently has excess capacity in the powder coating process, but adding additional capacity to this process is very costly. It also uses a great deal of space on the shop floor. The length of any of Chandler's products is limited to 60-foot-long steel sections, as this is the maximum size the powder coating equipment can accommodate.

SHIPPING AND INSTALLATION

Once the product parts are finished, they are packed for shipping. This requires some buffering and wrapping parts to protect the shape and finish. Products are generally shipped within a day or two of completion. Shipments occur on flat-bed trucks.

Products are installed on site by independent installers. Problems with installation can occur if the site was not properly prepared or described in the design process. Likewise, if the installer damages a part in the installation process, a new one must be manufactured. Chandler provides support for installation problems and guarantees its products. Shipping and installation costs are directly charged to the customer, so they are not included in Chandler's quoted prices.

COMPANY COSTS

Chandler utilizes a normal costing system—that is, each job is charged with its actual direct materials cost, and overhead is allocated to jobs based on cost drivers. Total overhead costs, drivers, and capacities can be found in Exhibit 1. Other costing information includes:

- Direct material is assigned to products at actual cost.
- Labor cannot be traced directly to individual jobs but is included in the cost of activities (such as design, engineering, fabrication, powder coating, scheduling/setup, and general factory).
- Design/drafting costs are assigned to jobs based on design hours.
- Engineering costs are assigned to jobs based on engineering hours.
- Scheduling costs include all setup costs. These are assigned to jobs based on the number of jobs being set up.
- Fabrication costs are assigned to jobs based on the number of parts.
- Powder coating costs are assigned to jobs based on square footage of the material.
- General factory costs include supervision, utilities, taxes, factory depreciation, and other factory costs. Chandler assigns general factory costs based on machine hours.
- Transportation costs are free on board (FOB) Chandler's facility. The customer is responsible for the shipping costs, which are paid directly to the trucking firm.

THE MEETING

Ned Beale: Dan, I got a call from a company in Florida that wants to sell one of our products at his swimming pool business as a pool shade. The interesting thing is, he wants 8 to 10 per month, indefinitely. This is a bit different from our usual order in that we would not have to go through the design and drafting or the engineering each time we get an order. Also, several units could be shipped out at one time, reducing shipping costs for the customer. There may be other economies of scale as well.

Dan Smith: You are right about the savings on drafting and engineering, but we need to consider our capacities. The additional business would put pressure on our workforce and processes, especially the powder coating system. As for the labor, we might be able to manage with overtime, or we may have to add another shift. Our current custom jobs are very profitable, and I do not want to make those wait while we produce multiple shades for this new customer.

Ned: While we are on the topic, last week I got a call from a landscaper who had a similar request for trellises. There are some savings we might discover by accepting these ongoing orders. Quoting prices, which we base on location, materials, square footage, and number of parts, would only have to be done once. Scheduling and setups could also be reduced if we produce in batches.

Dan: Do not commit to anything yet, but get an estimate of the resource usage and cost for each project, and I will take a look at them.

NEW OPPORTUNITIES

REQUEST FOR POOL SHADES IN FLORIDA

While on vacation, Mel Stutzman, manager of Luxury Pools in Ocala, Florida, noticed a shelter at a rest area along I-35 in Oklahoma. He thought this product would be a perfect complement to his pool business. Customers often asked about shade and shelter products for their pools and patio furniture. He found out this shelter was manufactured by Chandler Corporation and called the company as soon as he returned to inquire about ordering 120 of them for the coming year.

Initially Chandler would need to determine the adjustments to the structure to meet Florida's requirements due to hurricane-force winds and make minor design changes to accommodate the pools. Each subsequent structure would be identical. Details relating to materials and other resources needed for making the pool shades are in Exhibit 2.

REQUEST FOR LANDSCAPING TRELLISES

Jaime Potter saw one of Chandler's lattice-covered trellises and immediately knew this would be perfect for several of her clients. The lattice tops allow some sun to get through but provide enough shade to protect delicate plants from the full effect of the sun. Since the tops can be fabricated in many designs, she would not have to worry about having her designs all look alike. She placed a call to Chandler's production facility and is really hoping it will accept her order for 60 trellises in the next year. Details relating to materials and other resources needed for making the trellises are in Exhibit 2.

If Chandler decides to make either or both of the new products, it would be made one at a time, not as a batch, fitting them in between current orders when time is available, so as to not delay any of the custom orders. Current resource cost and capacities are in Exhibit 1. In the event that Chandler needs to purchase additional resources in order to make these new products, costs and capacities for new resources are in Exhibit 3.

REQUIREMENTS

Chandler Corporation is currently producing several products, including both custom and standard designs. It has received requests for two new products: swimming pool shades and landscape trellises. It is considering whether to produce either or both of these products. Based on the projections in Exhibits 1 to 3 describing the current capacity and required capacity needs for the new products, complete the following requirements:

1. **Target Costing:**
   - **a.** Determine the target cost of making a pool shelter given the price and desired profit margin shown in Exhibit 2.
   - **b.** Determine the target cost of making a trellis given the price and desired profit margin shown in Exhibit 2.
2. **Activity-Based Costing (ABC) and Capacity:**
   - **a.** Using ABC, find the current activity rates based on practical capacity.
   - **b.** Compute the cost of unused capacity using practical capacity rates.
   - **c.** Compute the total and unit costs of making the full demand of pool shades, assuming Chandler bases its rates on practical capacity. Also, assume Chandler would make these products one at a time when time is available, so as to not delay any of the custom orders.

d. Compute the total and unit costs of making the full demand of trellises, assuming Chandler bases its rates on practical capacity. Also, assume Chandler would make these products one at a time when time is available, so as to not delay any of the custom orders.

e. Compute the total and unit costs of making the full demand of pool shades, assuming Chandler bases its rates on expected capacity (used capacity plus required capacity for the full demand of pool shades). Also assume Chandler would make these products one at a time when time is available, so as to not delay any of the custom orders.

f. Compute the total and unit costs of making the full demand of trellises, assuming Chandler bases its rates on expected capacity (used capacity plus required capacity for the full demand of trellises). Also, assume Chandler would make these products one at a time when time is available, so as to not delay any of the custom orders.

g When considering a decision to make the new products, would costs computed using practical capacity (2c and 2d) or expected capacity (2e and 2f) as the denominator provide better information to Chandler's management? Explain your answer.

3. **Theory of Constraints:**

a. Assuming material is the only truly variable cost, find the throughput margin per unit for:

i. One pool shade

ii. One trellis

b. Assuming the powder coating process is the binding constraint and additional capacity is difficult or too costly to purchase, determine which product Chandler should be making first. Explain how and why you came to this conclusion. Also, show your computations to support your conclusion.

c. Assuming the powder coating process is the binding constraint and additional capacity is difficult or too costly to purchase, determine what mix of products should be made with the existing excess capacity.

d. Determine the incremental throughput margin Chandler would earn if it sold the mix you determined in 3c.

4. **Resource Spending Approach:**

Assume the decisions to make pool shades and trellises are considered to be a short-term decision and that Chandler would make these products one at a time when time is available, so as not to not delay any of the custom orders. Because there is excess capacity on current production equipment, the company wants to use a tactical decision (resource spending) approach to evaluate the decision to make the products. Exhibit 3 provides information about the cost to purchase additional resources.

a. Compute the net change in cash flow of making and selling the full demand of pool shades.

b. Compute the net change in cash flow of making and selling the full demand of trellises.

c. Compute the net change in cash flow of making the full demand for both the pool shades and the trellises.

d. Given the assumptions aforementioned (such as short-term and excess capacity), would Chandler's management choose to make either or both of the products? Explain your answer.

5. **Resource Usage Approach:**

Assume the decisions to make pool shades and trellises are considered to be a long-term but small-scale decision and that Chandler would make these products one at a time when time is available, so as not to not delay any of the custom orders. While still a tactical decision, excess capacity cannot be guaranteed beyond the current year. As such, Chandler feels a resource usage approach would be a more appropriate approach to evaluating the decision to make the products.

a. Compute the ABC rates for new resources.

b. Compute the incremental profit of making and selling the full demand of pool shades.

c. Compute the average cost of making one pool shade.

d. Compute the incremental profit of making and selling the full demand of trellises.

e. Compute the average cost of making one trellis.

f. Given the assumptions aforementioned (such as long-term and small-scale), would Chandler's management choose to make either or both of the products? Explain your answer.

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Exhibit 1. Existing Capacity Available and Used and Capacity Costs (Annual)

Resource	Practical Capacity	Cost Driver	Purchased Cost of Capacity	Used Capacity
Design/Drafting	20,000	Design hours	$500,000	19,500
Engineering	15,000	Engineering hours	$750,000	14,750
Fabrication	40,000	Number of parts	$30,000	39,000
Powder coating	400,000	Square feet	$1,800,000	210,000
Scheduling/setup	750	Job setups	$756,000	640
General factory	80,000	Machine hours	$2,400,000	64,000

Exhibit 2. Resource Needs for Potential New Products

Resource (Cost Level)	Cost Driver	Pool Shade Requirements	Trellis Requirements
Design/Drafting (product)	Design hours	30	20
Engineering (product)	Engineering hours	4	2
Fabrication (unit)	Number of parts	40	22
Powder coating (unit)	Square feet	1,600	170
Scheduling/setup (unit)	Job setups	1	1
General factory (unit)	Machine hours	80	10

	Pool Shade	Trellis
Selling price/unit	$42,500	$4,300
Material cost/unit	$18,750	$700

Desired Profit on Sales		0.30
Demand	Pool shades	120
Demand	Trellises	60

Exhibit 3. Purchase Price of Additional Capacity

Resource	Capacity of next step	Cost Driver	Cost of Next Step
Design/Drafting	2,000	Design hours	$50,000
Engineering	2,000	Engineering hours	$110,000
Fabrication	3,000	Parts	$3,150
Powder coating	20,000	Square feet	$100,000
Scheduling/setup	5,000	Job setups	$4,000
General factory	2,000	Machine hours	$36,000